

- **Capability (Tool):** A callable function or external service an agent can use.
- **Message:** Typed payload with headers for identity, causality, and policy decisions.
- **Policy:** Declarative rules that allow, transform, redact, or block actions.
- **Plan:** A structured decomposition of a goal into steps with dependencies.
- **Memory:** Context cache (short-term) and knowledge retrieval (long-term).
- **Orchestration:** Assignment, routing, and synchronisation across agents.
- **Evaluation:** Offline tests and online telemetry used to score quality, safety, and cost.

New features of the Microsoft Agentic Framework

The latest release of the Microsoft Agentic Framework introduces several enhancements aimed at improving scalability, intelligence, and developer productivity. Here are the most notable new features.

Enhanced multi-agent communication protocols: The framework now supports advanced messaging standards, allowing agents to communicate using secure, asynchronous channels. This improvement facilitates both real-time collaboration and reliable message delivery, even in distributed environments.

Plug-and-play agent modules: Developers can now integrate pre-built agent modules or create reusable components that can be plugged into existing solutions. This modular approach accelerates development and encourages best practices in code reuse.

Adaptive learning capabilities: Agents can be equipped with machine learning models that enable them to adapt their behaviours based on historical data and environmental feedback. These adaptive agents can refine their strategies over time, leading to smarter, more effective systems.

Visual orchestration designer: A new graphical interface allows

architects to design, configure, and monitor agent workflows visually, reducing the learning curve and improving transparency in complex solutions.

Robust monitoring and diagnostics:

The framework includes comprehensive tools for logging, real-time monitoring, and diagnostics, making it easier to identify bottlenecks, troubleshoot issues, and ensure system reliability.

Step-by-step guide: Utilising the new features of MAF for multi-agent development

Step 1: Setting up the framework

Install the Microsoft Agentic Framework via NuGet or the preferred package manager:

```
dotnet add package Microsoft.
AgenticFramework
```

Step 2: Defining agents and capabilities

Create a basic agent by inheriting from the AgentBase class and implementing the desired behaviours:

```
public class WeatherAgent : AgentBase
{
    public override Task
    OnMessageAsync(MessageContext context)
    {
        if (context.Message.Type ==
        "GetWeather")
        {
            var weatherInfo = FetchWeather(context.
            Message.Payload);
            return context.RespondAsync(new
            Message("WeatherResponse",
            weatherInfo));
        }
        return Task.CompletedTask;
    }
}
```

This agent responds to weather queries and demonstrates how agents can process messages asynchronously.

Step 3: Orchestrating agents visually

Launch the visual orchestration designer

provided with the framework. Drag and drop agent modules, define message flows, and set up triggers. This step allows you to design complex agent interactions without deep coding.

Step 4: Enabling adaptive learning

Integrate machine learning capabilities using the framework's built-in support:

```
public class AdaptiveAgent : AgentBase
{
    private readonly ILearningModule _
    learningModule;
    public AdaptiveAgent(ILearningModule
    learningModule)
    {
        _learningModule = learningModule;
    }
    public override async Task
    OnMessageAsync(MessageContext context)
    {
        var prediction = await _learningModule.
        PredictAsync(context.Message.Payload);
        await context.RespondAsync(new
        Message("PredictionResult",
        prediction));
    }
}
```

This approach allows agents to make data-driven decisions and evolve their responses over time.

Step 5: Monitoring and diagnostics

Leverage the built-in monitoring dashboard to track agent health, message throughput, and system performance. Set up alerts for anomalies and review diagnostic logs for troubleshooting.

This architecture is helpful in the following aspects.

- **Agility:** The framework's modular architecture and visual tools allow rapid prototyping and iteration, enabling teams to adapt solutions swiftly to changing requirements.
- **Extensibility:** With plug-and-play modules and support for third-party integrations, the framework accommodates evolving business